

$$1. \quad U_m(y) \propto \begin{cases} \cos\left(2\pi \frac{\sin \theta_m}{\lambda} y\right) & , m = 0, 2, 4 \\ \sin\left(2\pi \frac{\sin \theta_m}{\lambda} y\right) & , m = 1, 3, 4 \end{cases} \quad -\frac{d}{2} \leq y \leq \frac{d}{2} \quad U_m(y) \propto \begin{cases} e^{\delta m y} & , y > \frac{d}{2} \\ e^{-\delta m y} & , y < -\frac{d}{2} \end{cases}$$

a) Basta regular $y > 0$
continuidade em $y = \frac{d}{2}$

$m \rightarrow pm$

$$A_m \cos\left(\frac{\pi d}{\lambda} \sin \theta_m\right) = B_m e^{-\delta m \frac{d}{2}} \Rightarrow \frac{B_m}{A_m} \cos\left(\frac{\pi d}{2} \sin \theta_m\right) e^{\delta m \frac{d}{2}}$$

Para saber A_m usar normalização dos modos $\Rightarrow$ ver aula

$$b) \quad TE_0 (m=0) \quad n_1 = 1,48; \quad n_2 = 1,46; \quad d = 0,5 \mu m; \quad \lambda = 0,85 \mu m$$

Determinar as soluções $\sin \theta_m = x_m; \quad \Gamma = \frac{2d}{\lambda_0} \sqrt{n_1^2 - n_2^2} = 0,11 = 1$
 $\sin \theta_0 = x_0$

$$TE_0: \tan\left(\frac{\pi d}{\lambda} x_0\right) = \sqrt{\frac{1 - \frac{n_2^2}{n_1^2}}{\frac{n_2^2}{n_1^2}}} - 1$$

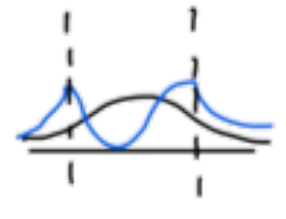
$$\left. \begin{aligned} \frac{\pi d}{\lambda} &= \frac{n_1 \pi d}{\lambda_0} = 2,735 \\ 1 - \frac{n_2^2}{n_1^2} &= 0,0268 \end{aligned} \right\} \tan(2,735 x_0) = \sqrt{\frac{0,0268}{\frac{\lambda_0^2}{\lambda_0^2}}} - 1 > 1 \Rightarrow x_0^2 < 0,0268 \Rightarrow x_0 < 0,164$$

$$\boxed{x_0 \approx 0,15}$$

$$U_0(y) = A_0 \cos\left(\frac{2\pi}{\lambda} x_0 y\right), \quad -\frac{d}{2} \leq y \leq \frac{d}{2}$$

$$U_0(y) = B_0 e^{\delta_0 y} = A_0 \cos\left(\frac{\pi d}{\lambda} x_0\right) e^{\delta_0 \frac{d}{2}} e^{-\delta_0 y}, \quad y > \frac{d}{2}$$

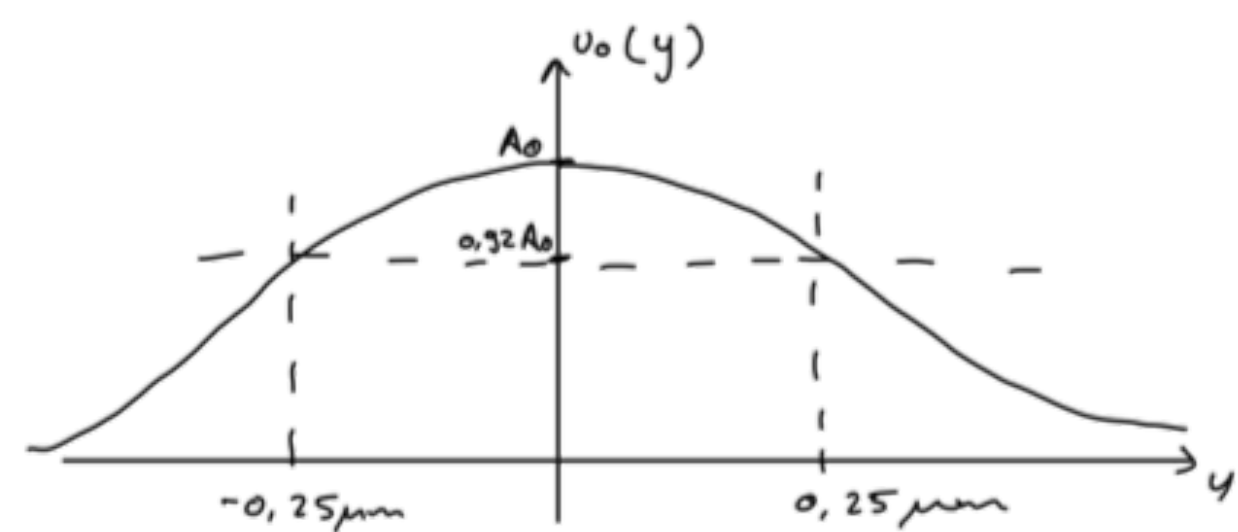
$$\delta_0 = n_2 k_0 \sqrt{\frac{\cos^2 \theta_0}{\frac{n_2^2}{n_1^2}} - 1} = n_2 k_0 \sqrt{\frac{1 - x_0^2}{\frac{n_2^2}{n_1^2}}} - 1 \Rightarrow \delta_0 = 0,72$$



Factor de distinção de campo na interface

$$\cos\left(\frac{\pi d}{\lambda} x_0\right) \approx 0,92 \quad e^{\delta_0 \frac{d}{2}} \approx 1,132 \quad \left. \begin{aligned} & \\ & \end{aligned} \right\} U_0(y) = A_0 \cos(1,64 y), \quad -0,25 \mu m \leq y \leq 0,25 \mu m$$

$$U_0(y) = A_0 1,101 e^{-0,72 y}, \quad y > 0,25$$



$$c) \quad T_0 = \frac{\int_0^{d/2} U_0^2(y) dy}{\int_0^{d/2} U_0^2(y) dy + \int_{d/2}^{\infty} U_0^2(y) dy}$$

$$\int_0^{0,25} A_0^2 \cos^2(1,64 y) dy = 0,236 A_0^2$$

$$\int_{0,25}^{\infty} 1,101^2 A_0^2 e^{-0,72 y} dy = 1,28 A_0^2$$

$$T_0 = \frac{0,236}{0,236 A_0^2 + 1,28 A_0^2} = 0,156 \approx 16\%$$

$$2. \quad U_m(y) = \begin{cases} A \cos(k_y y + \varphi) & , -\frac{d}{2} \leq y \leq \frac{d}{2} \\ B e^{\delta y} & , y > \frac{d}{2} \\ B e^{-\delta y} & , y < -\frac{d}{2} \end{cases} \quad E \times(y, z) = U(y) e^{-i\beta z}$$

$$a) \quad H_y(y, z) \text{ e } H_z(y, z)$$

$$\vec{\nabla} \times \vec{E} = -i\omega \mu \vec{H}$$

$$\vec{\nabla} \times \vec{E} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \partial_x & \partial_y & \partial_z \\ E_x & 0 & 0 \end{vmatrix} = [\partial_z E_x \vec{e}_y - \partial_y E_x \vec{e}_z] = -i\omega \mu (H_y \vec{e}_y + H_z \vec{e}_z)$$

Plano TE $\rightarrow$ campo elétrico só em x
TEM $\rightarrow$ campo elétrico em y e z

$$H_y = \frac{1}{-i\omega \mu} U(y) \partial_z e^{-i\beta z} \Rightarrow H_y = \frac{\beta}{\omega \mu} U(y) e^{-i\beta z}$$

$$\left. \begin{aligned} \text{Núcleo} \\ H_y &= \frac{\beta}{\omega \mu} A \cos(k_y y + \varphi) e^{-i\beta z} \\ H_z &= i \frac{k_y}{\omega \mu} A \sin(k_y y + \varphi) e^{-i\beta z} \end{aligned} \right| \begin{aligned} \text{Revestimento} \quad y > d/2 \\ H_y &= \frac{\beta}{\omega \mu} B e^{\delta y} e^{-i\beta z} \\ H_z &= i \frac{\delta}{\omega \mu} B e^{\delta y} e^{-i\beta z} \end{aligned}$$

b) Continuidade do campo elétrico

$$A \cos(k_y \frac{d}{2} + \varphi) = B e^{-\delta \frac{d}{2}} \quad (1)$$

Continuidade do campo magnético

$$i \frac{k_y}{\omega \mu} A \sin(k_y \frac{d}{2} + \varphi) e^{-i\beta z} = i \frac{\delta}{\omega \mu} B e^{\delta \frac{d}{2}} e^{-i\beta z} \Rightarrow k_y A \sin(k_y \frac{d}{2} + \varphi) = \delta B e^{\delta \frac{d}{2}} \quad (2)$$

Dividir (2)/(1)

$$\frac{k_y A \sin(k_y \frac{d}{2} + \varphi)}{A \cos(k_y \frac{d}{2} + \varphi)} = \frac{\delta B e^{\delta \frac{d}{2}}}{B e^{-\delta \frac{d}{2}}} \Rightarrow k_y \tan(k_y \frac{d}{2} + \varphi) = \delta \quad (*)$$

$$k_y = n_1 k_0 \sin \theta \quad \delta = n_2 k_0 \sqrt{\frac{\cos^2 \theta}{\cos^2 \theta_c} - 1}$$

$$\frac{\delta}{k_y} = \frac{n_2 k_0}{n_1 k_0} \frac{1}{\sin \theta} \sqrt{\frac{\cos^2 \theta}{\cos^2 \theta_c} - 1} = \frac{\cos \theta_c}{\sin \theta} \sqrt{\frac{\cos^2 \theta}{\cos^2 \theta_c} - 1} = \frac{1}{\sin \theta} \sqrt{\cos^2 \theta - \cos^2 \theta_c} = \sqrt{\frac{1 - \sin^2 \theta_c}{\sin^2 \theta} - 1} = \sqrt{-1 + \frac{\sin^2 \theta_c}{\sin^2 \theta}} = \sqrt{\frac{\sin^2 \theta_c}{\sin^2 \theta} - 1}$$

$$k_y \frac{d}{2} = n_1 k_0 \sin \theta \frac{d}{2} = n_1 \frac{2\pi}{\lambda_0} \sin \theta \frac{d}{2} = \frac{\pi d}{\lambda} \sin \theta$$

$$U(y) = A \cos(k_y y + \varphi) = A \cos\left(\frac{2\pi}{\lambda} \sin \theta y + \varphi\right) \begin{cases} A \cos\left(\frac{2\pi}{\lambda} \sin \theta y\right), \text{ modos pares} \\ A \sin\left(\frac{2\pi}{\lambda} \sin \theta y\right), \text{ modos ímpares} \end{cases}$$

$$\varphi = m \frac{\pi}{2}, \quad m = 0, 1, 2, \dots$$

$$(*) \Rightarrow \tan(k_y \frac{d}{2} + \varphi) = \sqrt{\frac{\sin^2 \theta_c}{\sin^2 \theta} - 1} \Rightarrow \tan\left(\frac{\pi d}{\lambda} \sin \theta + \varphi\right) = \sqrt{\frac{\sin^2 \theta_c}{\sin^2 \theta} - 1}$$

$$\Rightarrow \tan\left(\frac{\pi d}{\lambda} \sin \theta + m \frac{\pi}{2}\right) = \sqrt{\frac{\sin^2 \theta_c}{\sin^2 \theta} - 1}$$

$$3.a) \quad TE_{01} \quad a = 0,5 \mu m$$

Acoplamento ótico

guia simétrico $\Rightarrow C_{21} = C_{12}$

$$C_{21} = \frac{1}{2} (n_2^2 - n_1^2) \frac{k_0^2}{\beta_1} \int_a^{a+d} dy U_1(y) U_2(y) = 0,01 \mu m^{-1}$$

Regime de acoplamento para os campos são tratados como se não existisse a segunda guia

$$I = A_0^2 \cos^2\left(\frac{\pi d}{\lambda} x_0\right) e^{\delta_0 \frac{d}{2}} \int_{0,5}^{10} dy \cos^2(1,64 y) e^{-0,72 y} = 0,115 A_0^2$$

Ver expressão para A_m
 $A_0^2 = 9,69$

$$b) \quad L_0 = \frac{\pi}{2\epsilon} \Rightarrow L_0 = 157,1 \mu m \rightarrow 3dB - coupler \quad \frac{L_0}{2} = 78,55 \mu m$$